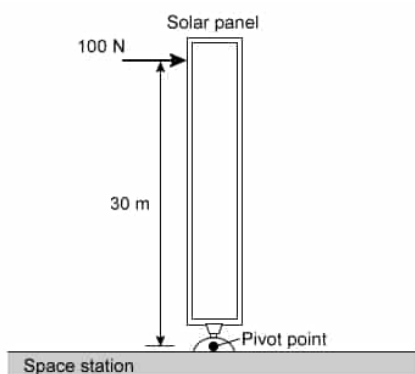


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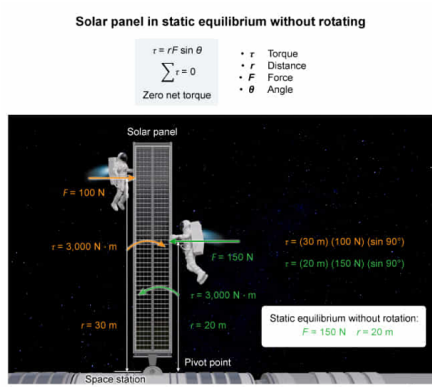
Two astronauts use their jet packs to position a solar panel attached to a large space station by a pivot point, as shown above. The solar panel is at rest relative to the space station and not rotating. One astronaut applies 100 N of force 30 m from the pivot point toward the right. What is the force applied toward the left by the second astronaut, and how far from the pivot point is it applied to prevent the solar panel from rotating?

- ☐ A. 100 N force applied 10 m from the pivot point [4%]
- ☐ B. 100 N force applied 20 m from the pivot point [6%]
- ☐ C. 150 N force applied 10 m from the pivot point [8%]
- ☒ D. 150 N force applied 20 m from the pivot point [80%]

Correct

79% answered correctly

Explanation:



An object in **static equilibrium** is at rest, and all **forces** F acting on it sum to zero:

$$\sum F = 0$$

Furthermore, static equilibrium implies that the object is not rotating. Thus, all **torques** τ acting on the object must also sum to zero:

$$\sum \tau = 0$$

The τ equals the product of the **distance** r from the axis of rotation, the magnitude of F , and the sine of the angle θ between r and F :

$$\tau = rF \sin \theta$$

By convention, a positive τ causes the object to spin **counterclockwise**, and a negative τ spins the object **clockwise**. Typically, the object can rotate around its center of mass or around a fixed point, like a pivot point.

In this question, one astronaut applies a F of 100 N perpendicular to the solar panel at a r of 30 m from the pivot point. This induces a τ on the solar panel equal to:

$$\tau = (30 \text{ m})(100 \text{ N})(\sin 90^\circ) = 3,000 \text{ N} \cdot \text{m}$$

Because this τ causes the solar panel to rotate clockwise, it is considered negative.

To remain in static equilibrium, the second astronaut must induce a positive τ of 3,000 N·m. Applying a F of 150 N perpendicular to the solar panel at a r of 20 m toward the left induces a positive τ of:

$$\tau = (20 \text{ m})(150 \text{ N})(\sin 90^\circ) = 3,000 \text{ N} \cdot \text{m}$$

Therefore, the solar panel remains in static equilibrium.

(Choices A and B) A force of 100 N can only induce a torque of 3,000 N·m if it is applied at a distance of 30 m from the pivot point, not 10 m or 20 m.

(Choice C) Applying a 150 N force at a distance of 10 m from the pivot point induces a torque of 1,500 N·m on the solar panel, which is less than the 3,000 N·m torque applied by the first astronaut.

Educational objective:

For an object in static equilibrium, the sum of all torques exerted on the object equals zero. The torque on an object is equal to the product of the distance from the pivot point that the force is applied, the magnitude of the force, and the sine of the angle between the distance and the force.

Time spent: 0 sec

Subject:

Physics

QID: 404037

System:

Mechanics and Energy

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An astronaut on Mars uses a short burst from a rocket engine to launch an atmosphere sampler vertically upward to different heights. If the speed of the rocket at the end of the engine burst were doubled, then the maximum height of the rocket would:

- ✓ ☒ A. increase by a factor of 4. [61%]
☐ B. increase by a factor of 2. [32%]
☐ C. decrease by a factor of 2. [3%]
☐ D. decrease by a factor of 4. [2%]

Correct

62% answered correctly

Explanation:

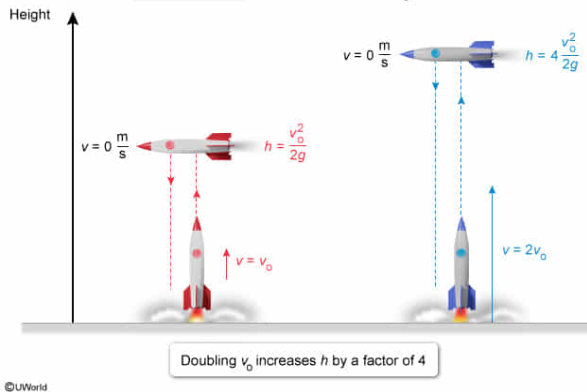
Height of a rocket from initial speed

$$v^2 = v_0^2 + 2a\Delta x$$

$$v = 0 \quad a = -g$$

$$h = \frac{v_0^2}{2g}$$

- v Speed
- v_0 Initial speed
- a Acceleration
- Δx Displacement
- g Gravitational acceleration
- h Maximum height



The motion of a projectile follows the [translational motion equations](#). In particular, the velocity v of a projectile during constant [acceleration](#) a is given by:

$$v^2 = v_0^2 + 2a \Delta x$$

where v_0 is the initial velocity and Δx is the displacement.

For a projectile traveling straight upward, Δx corresponds to its height above the ground. Furthermore, the projectile is subjected to the a of **gravity** g , which is negative because it acts in the opposite direction of the projectile's motion. At its **maximum height** h , the projectile's v equals zero and it begins to fall back down. Hence, the above equation when the projectile is at h simplifies to:

$$0 = v_0^2 + 2(-g)h$$

Solving this equation for h yields:

$$h = \frac{v_0^2}{2g}$$

In this question, the v_0 of the rocket is doubled, giving a new maximum height h_{new} of:

$$h_{\text{new}} = \frac{(2v_0)^2}{2g} = 4\left(\frac{v_0^2}{2g}\right)$$

Therefore, the height of the rocket increases by a factor of four when the initial velocity is doubled.

(Choice B) The maximum height of the rocket does not increase by a factor of two when v_0 doubles because h depends on the square of v_0 .

(Choices C and D) The maximum height of the rocket increases, not decreases,

when the initial velocity increases.

Educational objective:

Translational motion equations are used to calculate the height reached by a launched projectile. The maximum height of a projectile's path is proportional to the square of the initial velocity.

Time spent:0 sec

Subject:
Physics

QID:404038

System:
Mechanics and Energy

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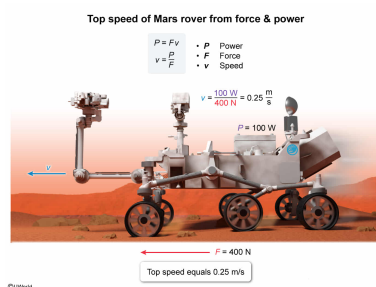
An emergency during a space flight could require astronauts to leave the spacecraft to perform repairs. A space suit provides a livable environment for the astronaut, but controlled movement in space is difficult due to the relative lack of gravitational attraction. Astronauts can use jet packs that release short bursts of compressed gas in different directions, allowing them to control their movements in open space outside the spacecraft.

At its top speed, a Mars rover generates a constant force of 400 N and uses 100 W of battery power. What is the top speed of the rover?

- ☐ A. 2.5 m/s [14%]
- ☐ B. 1.0 m/s [12%]
- ☒ C. 0.25 m/s [72%]
- ☐ D. 0.10 m/s [1%]

Correct
72% answered correctly

Explanation:



For an object to maintain a **constant speed** v , it requires a constant **force** F to be applied, which counteracts any phenomena that act to slow it down, such as **friction** or **air resistance**. The **power** P required to maintain a constant speed v equals the product of F and v :

$$P = Fv$$

This equation can be rearranged to solve for v , giving:

$$v = \frac{P}{F}$$

In this question, the rover generates a F of 400 N and uses 100 W of P from its batteries when traveling at its top speed. Substituting these values into the above equation yields:

$$v = \frac{100 \text{ W}}{400 \text{ N}} = 0.25 \frac{\frac{\text{kg} \cdot \text{m}^2}{\text{s}^3}}{\frac{\text{kg} \cdot \text{m}}{\text{s}^2}} = 0.25 \frac{\text{m}}{\text{s}}$$

Therefore, the top speed of the rover is 0.25 m/s.

(Choice A) 2.5 m/s is calculated by incorrectly including the force of gravitational acceleration on Earth (10 m/s^2) in the equation.

(Choice B) 1.0 m/s results from incorrectly multiplying by the gravitational acceleration on Mars (4 m/s^2).

(Choice D) 0.10 m/s is obtained by incorrectly calculating the ratio of gravitational acceleration on Earth and the battery power of the rover.

Educational objective:

The power required for an object to maintain a constant speed equals the product of the applied force and its speed. A constant force must be applied to overcome the forces acting to slow down the object, such as friction and air resistance.

Time spent:0 sec
Subject:
Physics

QID:404039
System:
Mechanics and Energy

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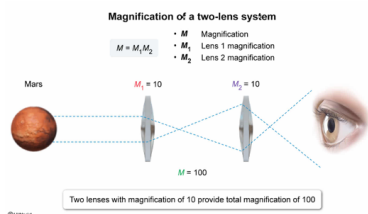
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A researcher builds a telescope with a magnification of 100 to view Mars from Earth using lenses with magnifications of either 10 or 25. To achieve the desired magnification, which of the following lenses were combined?

- ✓ ☒ A. Two lenses with magnification of 10 [68%]
☐ B. Two lenses with magnification of 25 [5%]
☐ C. Four lenses with magnification of 25 [21%]
☐ D. Two lenses with magnifications of 10 and 25 [3%]

Correct
68% answered correctly

Explanation:



Lenses provide magnification M equal to the ratio of the distances from the lens to the image i and the lens to the object o :

$$M = \frac{i}{o}$$

Multiple lenses can be combined into one system, resulting in a final M equal to the **product** of the magnifications M_n of each individual lens:

$$M = M_1 M_2 M_3 \dots$$

In this question, a researcher uses lenses with M_n equal to either 10 or 25 to build a telescope with a final M equal to 100. Combining two lenses with M_1 equal to 10 and M_2 equal to 10 yields:

$$M = M_1 M_2 = 10 \cdot 10 = 100$$

Therefore, the researcher combined two lenses with M_n equal to 10 to make a telescope with a final M of 100.

(Choice B) Combining two lenses each with a magnification of 25 provides a total magnification of $25 \cdot 25 = 625$, not 100.

(Choice C) Combining four lenses with magnifications of 25 will not produce a total magnification of 100 because the individual magnifications are multiplied together, not summed.

(Choice D) Using lenses with magnifications of 10 and 25 would yield a telescope with a total magnification of 250, not 100.

Educational objective:

The total magnification for a system of lenses equals the product of the magnification of each individual lens.

Time spent: 0 sec
Subject:
Physics

QID: 404040
System:
Mechanics and Energy

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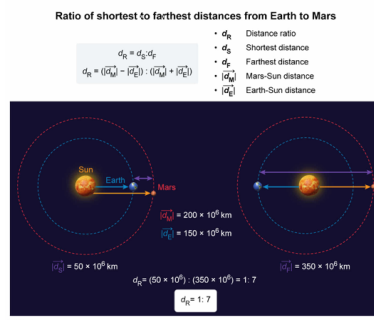
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What is the ratio of the shortest to farthest distances between Earth and Mars as these planets orbit the Sun?

- ☐ A. 3:4 [42%]
☐ B. 1:3 [10%]
☐ C. 1:4 [11%]
☒ D. 1:7 [35%]

Correct
 35% answered correctly

Explanation:



As a planet orbits a star, its position $\vec{d}$ relative to its star can be approximated as a **vector** with a constant **magnitude** $|\vec{d}|$ and a constantly changing **direction** as the planet moves around the star. Two planets are closest to each other when their position vectors align on the same side of their star. Conversely, the planets are farthest from each other when their position vectors are in opposite directions.

Given the distances $|\vec{d}_1|$ and $|\vec{d}_2|$ of two planets from the star they both orbit, the shortest distance d_S between the planets is the difference of $|\vec{d}_1|$ and $|\vec{d}_2|$, and the farthest distance d_F between them is the sum of $|\vec{d}_1|$ and $|\vec{d}_2|$.

In this question, the passage gives the distance $|\vec{d}_M|$ from Mars to the Sun as 200 million km and the distance $|\vec{d}_E|$ from Earth to the Sun as 150 million km. Hence, d_S equals:

$$d_S = |\vec{d}_M| - |\vec{d}_E| = (200 \times 10^6 \text{ km}) - (150 \times 10^6 \text{ km}) = 50 \times 10^6 \text{ km}$$

Similarly, d_F equals:

$$d_F = |\vec{d}_M| + |\vec{d}_E| = (200 \times 10^6 \text{ km}) + (150 \times 10^6 \text{ km}) = 350 \times 10^6 \text{ km}$$

The ratio d_R of d_S and d_F equals:

$$d_R = d_S : d_F = (50 \times 10^6) : (350 \times 10^6) = 1:7$$

Therefore, the ratio of the shortest and farthest distances between the Earth and Mars is 1:7.

(Choice A) 3:4 is the ratio of the average distances from Earth and Mars to the Sun, not the ratio of the shortest and farthest distances between them.

(Choice B) 1:3 is the ratio of the shortest distance between the two planets and the distance from Earth to the Sun.

(Choice C) 1:4 is the ratio of the shortest distance between Earth and Mars and the distance from Mars to the Sun, not the ratio of the planets' shortest and farthest distances.

Educational objective:

The positions of two objects circling a central point can be represented by two vectors. The objects are nearest to each other when their position vectors align in the same direction, and farthest away from each other when their position vectors are in opposite directions.

Time spent:0 sec
Subject:
Physics

QID:404041
System:
Mechanics and Energy

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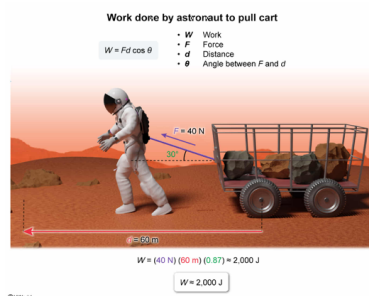
An astronaut pulls a cart of rock samples a distance of 60 m along flat ground using a tether attached to their spacesuit. The tether has a tension of 40 N and an angle of 30° relative to the horizontal. What is the approximate work the astronaut does on the cart? (Note: $\sin 30^\circ = 0.50$; $\cos 30^\circ = 0.87$)

- ☐ A. 1,200 J [18%]
☒ B. 2,000 J [62%]
☐ C. 2,400 J [14%]
☐ D. 2,800 J [3%]

Correct

62% answered correctly

Explanation:



The **work** W required to move an object is the product of the **force** F applied, the magnitude of the object's **displacement** d , and the cosine of the **angle** θ between F and d :

$$W = Fd \cos \theta$$

In this question, the astronaut pulls the cart along flat ground using a tether attached to their spacesuit. Hence, the **tension** T on the tether represents the F the astronaut applies to the cart:

$$W = Td \cos \theta$$

Furthermore, θ equals 30° because d is in the horizontal direction. Substituting the values into the above equation yields:

$$W = (40 \text{ N})(60 \text{ m})(0.87) \approx 2,000 \text{ J}$$

Therefore, the astronaut does approximately 2,000 J of work on the cart.

(Choice A) 1,200 J is calculated by using $\sin \theta$ in the work equation instead of $\cos \theta$.

(Choice C) 2,400 J results from omitting the $\cos \theta$ term in the work equation.

(Choice D) 2,800 J is obtained by using the inverse of $\cos \theta$ in the work equation.

Educational objective:

The work done to pull a cart on flat ground equals the product of the applied force, the displacement, and the cosine of the angle of the force relative to the horizontal.

Time spent: 0 sec
Subject:
Physics

QID: 404042
System:
Mechanics and Energy